Using MedGen to assign clinical categories to tests and utilize the disease categories as a search criterion for tests in the NIH Genetic Testing Registry (GTR®)

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MedGen (https://www.ncbi.nlm.nih.gov/medgen) integrates multiple clinically relevant and sometimes orthogonal ontologies and vocabularies under one resource. The source ontologies in MedGen include the Human Phenotype Ontology (HPO), Mondo, and OrphaNet; vocabularies include a subset of the Unified Medical Language System (UMLS), Online Mendelian Inheritance in Man (OMIM), PharmGKB, GeneReviews, and NCATS' Genetics and Rare Disease (GARD) dataset. Additionally, to support submissions to NCBI products such as the NIH Genetic Testing Registry (GTR®) and ClinVar, novel terms are created and assigned an unique identifier. MedGen source data are integrated using equivalent concepts identified between sources and mapped to a single MedGen record that can be searched by concept names, synonyms or identifiers. Automated mapping of terminologies uses identifier mappings from the source when available or customized string mapping algorithms. When the automated pipelines find discrepancies in mapping concepts across vocabularies, the terms are assigned to curators for review. When applicable, these data discrepancies are sent to the sources, so data is standardized across the community supporting interoperability efforts. The subset of UMLS concepts in MedGen including SNOMED CT, MeSH, and NCI thesaurus have significant overlap with HPO and OMIM, while Mondo and PharmGKB have fewer overlaps.

The NIH Genetic Testing Registry (GTR®, https://www.ncbi.nlm.nih.gov/gtr) is a database of clinical and research genetic tests offered by participating laboratories. Each test is accessioned and versioned, and data is submitted by the testing laboratories. Test descriptions include the test name, indication (mapped to a disease concept from MedGen), test targets, methodology, clinical and analytical validity, and clinical utility. In order to support clinicians' selection of the most suitable test for their patients, we devised a way for a test to be identified by the broad clinical specialty(ies) a test falls under by looking at the disease(s) listed on the test. Our method provides clinical category or medical specialty labels for tests. This work used data and concept hierarchies from HPO

and Mondo. GTR and MedGen are housed at the National Center for Biotechnology Information (NCBI) at the National Library of Medicine (NLM).

To create clinical specialties for tests, we first create clinical category labels and then assign match a higher-level nodes HPO and/or Mondo nodes to each label. Any descendant or child of the assigned top node inherits the clinical category label. The submitted indications or diseases on tests are usually the terminal nodes of the ontologies. A GTR test has at least one disease but can have multiple diseases. Some tests including large panels, exomes and genomes can have hundreds of diseases. To get the clinical category labels for tests, the labels for all the indicated diseases are collected and scored. The clinical category label that is most represented gets the top score , the one that is least represented gets the lowest score, and the rest are scored in between. The scores are used to prioritize the categories for display on a test page. We have assigned clinical categories to tests in GTR, leveraging HPO and Mondo ontology hierarchies and indications described in the test. Using the calculated clinical categories, we have enabled retrieval of tests that fit a broad clinical category in the search functionality in GTR's advanced search.

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